

# Lab 02

## Input Output Statement

# 1

## Review

### 1.1. Programming

It is the process to write a program / an application.

Specifically, programming includes 4 steps:

- Step1: Understanding your problem.
- Step 2: Designing an algorithm.
- Step 3: Writing source code.
- Step 4: Packaging source code files into an executable file, a website, an apk, or an ipa file...

In this lab, we practice to write source code in C++.

### 1.2. Input and Output

- Basic units: variables/constants, data-types, expressions.
- Input output stream: concepts, stdin/stdout, formatting.
- Text file: file stream, read/write.

# 2

## Reading

### 2.1 Online IDE

Study how to write, run, enter input values and share source codes in *ideone.com*.

Find other 2 similar websites.

### 2.2 Compiling program

Study and answer the following questions:

- a) How to compile a C/C++ program?  
<https://www.geeksforgeeks.org/compiling-a-c-program-behind-the-scenes/>
- b) Study at least 2 common C/C++ compilers.
  - <https://gcc.gnu.org/onlinedocs/gcc/>
  - <https://clang.llvm.org/docs/UsersManual.html>
- c) Install and compile 3 program in Assignment 1 in gcc compiler.
  - <https://aroliant.github.io/guides/installation/windows/gcc.html>
  - [https://youtu.be/hSAY50OL\\_J4](https://youtu.be/hSAY50OL_J4)

# 3

## Requirements

**In-class assignment: P10, P15, P13.**

**Homework: P10, P12, P13, P15, P16, P17, P18, P23, P24, P25.**

(Students need to check 'W01-Problems.pdf' file for Px)

Students must design at least 3 test cases for each program.

```
// <YOUR STUDENT ID>
// <YOUR FULL NAME>
// <YOUR CLASS>
```

```
// Test case 1
// Input:
// Output:
```

```
// Test case 2
// Input:
// Output:
```

```
// Test case 3
// Input:
// Output:
```

```
int main(){
    //your code here

    return 0;
}
```

### 3.1. P02

Write a program that inputs two integers. Calculate the sum of these two integers and print the result on the screen.

**Input data:**

A single line containing two integers a and b, separated by a space.

Where  $-10^9 \leq a, b \leq 10^9$ .

**Output data:**

The sum of the two integers, in the format  $a + b = c$ .

**Example:**

| Input | Output |
|-------|--------|
| 3 5   | 3+5=8  |

### 3.2. P05

Write a program that inputs two integers. Divide the first number by the second and print the result on the screen.

**Input data:**

A single line containing two integers a and b, separated by a space.

Where  $-10^9 \leq a, b \leq 10^9$ .

**Output data:**

The result is in the format  $a / b = c$ , rounded to 2 decimal places.

**Example:**

| Input | Output       |
|-------|--------------|
| 9 2   | 9 / 2 = 4.50 |

### 3.3. P13

Write a program that inputs the lengths of the three sides of a valid triangle. Calculate the perimeter and area of that triangle and print them on the screen.

**Input data:**

A single line containing three positive real numbers, edge1, edge2, edge3, representing the lengths of the three sides of the triangle.

Where  $0 < \text{edge1}, \text{edge2}, \text{edge3} \leq 10^9$ .

**Output data:**

The perimeter and area of the triangle on the same line, rounded to two decimal places.

**Example:**

| Input       | Output     |
|-------------|------------|
| 3.0 4.0 5.0 | 12.00 6.00 |

### 3.4. P10

Write a program that inputs the electricity meter reading from the previous month and the current month. Calculate the number of KWh we have consumed and print the result on the screen.

**Input data:**

A single line containing two positive integers, previous and current, which are the electricity meter readings for the previous and current months, separated by a space.

Where  $0 \leq \text{previous} \leq \text{current} \leq 10^9$ .

**Output data:**

The consumed electricity reading.

**Example:**

| Input     | Output |
|-----------|--------|
| 1000 1211 | 211    |

### 3.5. P12

Write a program that inputs the quantity and unit price of a product.

Calculate the total amount to be paid = cost of goods + tax.

Cost of goods = quantity \* unit price.

Tax = 10% of the cost of goods.

**Input data:**

Line 1: a positive integer, quantity, representing the quantity of the product.

Line 2: a positive real number, price, representing the unit price of the product.

Where  $-10^9 \leq \text{quantity}, \text{price} \leq 10^9$ .

**Output data:**

The total amount to be paid, with 0 decimal places.

**Example:**

| Input      | Output |
|------------|--------|
| 7<br>30000 | 231000 |

### 3.6. P16

Write a program that inputs a Vehicle registration plate, which is a positive integer with 5 digits. Calculate the last digit of the sum of the digits (called "nut") of that license plate number

**Input data:**

A positive integer plate with 5 digits, where  $10000 \leq \text{plate} \leq 99999$ .

**Output data:**

The sum of the digits (nut).

**Example:**

| Input | Output |
|-------|--------|
| 12345 | 5      |

### 3.7. P17

Write a program that inputs a positive integer money, which is an even number in the thousands. Consider the following banknote denominations: 500.000; 200.000; 100.000; 50.000; 20.000; 10.000; 5000; 2000; and 1000. Using a method that prioritizes higher denominations first, print the number of each banknote needed for the given amount.

**Input:**

A single positive integer money, representing the amount to be exchanged, where  $0 < \text{money} \leq 10^9$ .

**Output:**

9 lines, each indicating the denomination and the number of notes of that denomination, in the format: denomination: number of notes.

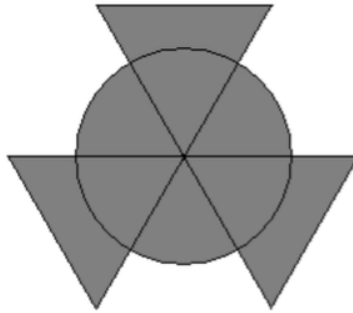
**Example:**

| Input   | Output                                                                                                         |
|---------|----------------------------------------------------------------------------------------------------------------|
| 2361000 | 500.000: 4<br>200.000: 1<br>100.000: 1<br>50.000: 1<br>20.000: 0<br>10.000: 1<br>5000: 0<br>2000: 0<br>1000: 0 |

### 3.8. P23

Write a program that inputs the side length of an equilateral triangle and the radius of a circle. Calculate the total area of the gray region.

Use the convention  $\pi = 3.14$

**Input Data:**

A single line containing 2 positive real numbers, edge and radius, separated by a space.

Where  $0 < \text{edge}, \text{radius} \leq 10^9$ .

**Output Data:**

A single number, area, representing the area of the shaded region, rounded to 2 decimal places.

**Example:**

| Input | Output |
|-------|--------|
| 3 2   | 10.17  |

### 3.9. P24

A promotional program offers a 40% cashback on all transactions, with a maximum cashback limit of 100,000 VND. Determine how much a user should spend to receive the maximum cashback amount.

**Input Data:**

A single line containing 2 positive real numbers, percent and quota, separated by a space.

Where  $0 < \text{percent}, \text{quota} \leq 10^9$ .

**Output Data:**

A single number representing the amount to spend to receive the maximum cashback, rounded to 2 decimal places.

**Example:**

| Input     | Output |
|-----------|--------|
| 40 100000 | 250000 |

### 3.10. P25

On 12-10-2019, Eliud Kipchoge completed 42.195 km in 1 hour, 59 minutes, and 40.2 seconds, becoming the first person to run a marathon distance of 42 km in under 2 hours.

Write a program to calculate the pace (minutes per km) and speed (km/h) with given the distance (km) and the running time (hours, minutes, seconds).

**Input Data:**

A single line containing 4 positive real numbers: km, hour, minute, and second, separated by spaces.

Where  $0 < \text{km, hour, minute, second} \leq 10^9$ .

**Output Data:**

A single line with two numbers representing the pace and speed, each rounded to 2 decimal places.

**Example:**

| Input            | Output     |
|------------------|------------|
| 42.195 1 59 40.2 | 21.16 2.84 |